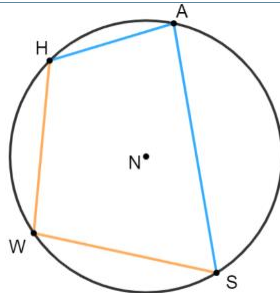


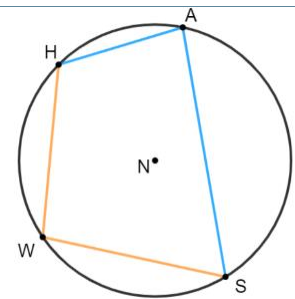
11.4 Solving Problems Involving Quadrilaterals Inscribed in Circles

Lesson Introduction

 Benchmark Focus	MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.		 Learning Targets	I can solve mathematical problems involving quadrilaterals inscribed in a circle.
 Benchmark Coherence	Building On			Working Toward
	MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.			N/A
 Essential Questions	- How do you solve mathematical problems involving quadrilaterals inscribed in a circle? - What is the relationship between the opposite angles in quadrilaterals inscribed in circles?			
 Lesson Narrative	In this lesson, students decompose quadrilaterals inscribed in circles into two triangles. They use the relationship between inscribed angles and intercepted arcs to find unknown angles and arcs. Through guided and independent practice, students discover that the opposite angles in quadrilaterals inscribed in circles are supplementary.			
 Component Summary	11.4.1 Warm-Up (~5 min.)	Notice and Wonder about Machu Picchu (<i>SE p. 197</i>)	11.4.3 Guided Practice (10-15 min.)	Guided Practice finding angles in quadrilaterals inscribed in circles (<i>SE p. 199</i>)
	11.4.2 Exploration (15 min.)	Make conjectures about angles in quadrilaterals inscribed in circles (<i>SE p. 198</i>)	11.4.4 Your Turn! (10 min.)	Independent practice finding angles in quadrilaterals inscribed in circles (<i>SE p. 200</i>)
	11.4.5 Wrap-Up (~5 Min)	Identify and correct errors in sample student work (<i>SE p. 201</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Inscribed Angle - Intercepted Arc - Supplementary		(Optional) Colored Pencils	Prior to the lesson, work through 11.4.2 Exploration, 11.4.3 Guided Practice, and 11.4.4 Your Turn! to ensure that you are familiar with the steps students will take and are prepared to support them as they work.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not understand the relationship between the inscribed angle and the intercepted arc. - Students may struggle with identifying opposite angles. - Students may solve for opposite angles using congruency instead of supplementary.	- Consider displaying the anchor chart from Lesson 3 with the relationship between the inscribed angle and the intercepted arc for students to reference. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the relationship between the opposite angles in quadrilaterals inscribed in circles.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder During 11.4.1 Warm Up, prompt students to reflect on what they notice and wonder about Machu Picchu. Call on a variety of students to share what they noticed and wondered. This activity provides a low floor, as all students can participate, and high ceiling, as students can be challenged to think of multiple noticings and wonderings.	MA.K12.MTR.4.1: Discussion and Reflection In 11.4.5 Wrap-Up, students have the opportunity to demonstrate their mastery of the learning target by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors themselves in the future.
 Differentiate Instruction	Provide an auditory entry point in 11.4.2 Exploration by having students discuss their conjectures about the relationships of the angles in quadrilaterals inscribed in circles. Provide a visual entry point by prompting students to color code the opposite angles in the diagrams in 11.4.3 Guided Practice. Provide a kinesthetic entry point in 11.4.4 Your Turn! by having students construct their own diagram and create problems involving a quadrilateral inscribed in a circle.	
Lesson Notes:		





11.4.1 Warm-Up: Notice and Wonder

Component Overview: Students are engaged by a photograph and written description of Machu Picchu. Then, they are asked to share what they notice and wonder about Machu Picchu. This component exposes students to a historic real-world application, helping them see the utility of geometry.



11.4.1 Warm-Up: Notice and Wonder

Machu Picchu is an Inca citadel in southern Peru. Machu Picchu was built in the classical Inca style with polished dry-stone walls and consists of more than 150 buildings. Although many of the stones that were used to build the city were more than 50 pounds, it is believed that no wheels were used to transport these rocks up the mountain.



This activity provides students with an image of Machu Picchu, Peru. Students are asked to notate what they notice from the image and what they wonder from the image. This routine provides a low entry point for students to engage in general discussion.

1. What do you notice?

Answers will vary. I noticed that there are a lot of houses, which means a lot of carrying of heavy stones. I noticed that there is an open space in between buildings.

2. What do you wonder?

Answers will vary. I wonder how many people they had to carry the stones needed. I wonder if the materials were stored in the open green space.

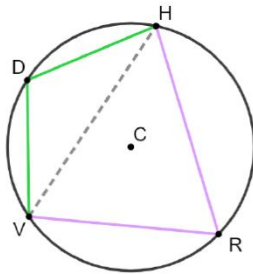
11.4.2 Exploration: Quadrilaterals Inscribed in Circles

Component Overview: Students make, discuss, and revise conjectures about angles in quadrilaterals inscribed in circles. Encourage students to support their conjectures with justifications and ask others if they agree or disagree and why.



11.4.2 Exploration: Quadrilaterals Inscribed in Circles

1. Circle C is shown, where $\overline{DV} \cong \overline{DH}$. $m\widehat{RV} = 100^\circ$, $m\widehat{DV} = 68^\circ$, $m\widehat{HDV} = 136^\circ$, and $m\widehat{HRV} = 224^\circ$.



- a. Complete the table.

Angle Measures	Angle Measures
$m\angle RHV = 50^\circ$	$m\angle HDV = 112^\circ$
$m\angle DHV = 34^\circ$	$m\angle DVH = 34^\circ$
$m\angle HRV = 68^\circ$	$m\angle HVR = 62^\circ$

- b. Work with your partner to make a conjecture about the relationships of the angles in $DHVR$.
Answers will vary. The opposite angles are supplementary.
- c. Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials
<i>Answers will vary</i>		
<i>Answers will vary</i>		



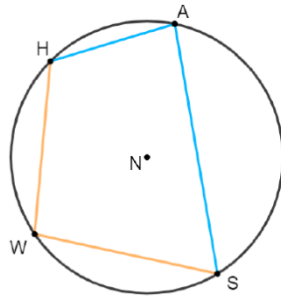
For this explanation, allow students to work with a partner to answer the scaffolded questions, rather than being guided through by a teacher. There will be an opportunity in the next component to summarize and synthesize what is discovered.



Consider displaying the anchor chart from Lesson 3 with the relationship between the inscribed angle and the intercepted arc for students to reference.

11.4.2 Exploration: Quadrilaterals Inscribed in Circles (continued)

- d. Review your conjecture with your partner. If necessary, make any changes.
2. Circle N has points H , A , S , and W on the circle. Quadrilateral $HASW$ is inscribed in the circle. $m\widehat{HA} = 59^\circ$, $m\widehat{HWS} = 166^\circ$, and $m\widehat{WHA} = 136^\circ$.



Use your conjecture to find the measures.

Angle Measures

$$\begin{aligned} m\angle WHA &= 112^\circ \\ m\angle HAS &= 83^\circ \\ m\angle ASW &= 68^\circ \\ m\angle SWH &= 97^\circ \end{aligned}$$

Arc Measures

$$\begin{aligned} m\widehat{SA} &= 135^\circ \\ m\widehat{SW} &= 89^\circ \\ m\widehat{HW} &= 77^\circ \end{aligned}$$



Some students may benefit from sentence stems or answer choices to help them form their conjectures.



Collect and Display

As students share their conjectures, collect a variety of their conjectures, and display them for all to see. Then, engage students in a discussion about whether they agree or disagree and why.

11.4.3 Guided Practice: Quadrilaterals Inscribed in Circles

Component Overview: Students engage in Guided Practice finding angles in quadrilaterals inscribed in circles.



11.4.3 Guided Practice: Quadrilaterals Inscribed in Circles

1. Complete the statement.

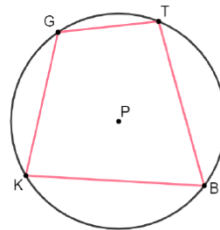
If a quadrilateral is inscribed in a circle,

- adjacent
- opposite

angles are

- congruent.
- complementary.
- supplementary.

2. Circle P has $GTBK$ inscribed in the circle. $m\widehat{TBK} = 206^\circ$ and $m\angle GKB = 84^\circ$.



- a. What is the $m\angle TBK$?
 77°
- b. What is the $m\angle GTB$?
 96°



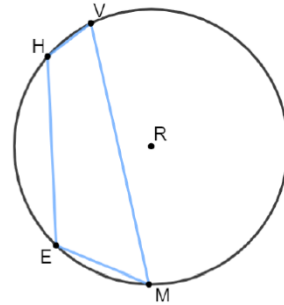
Check for understanding after question 1 to make sure that students understand the relationship between the angles in quadrilaterals inscribed in circles. Use this data to determine how much guidance is needed in this Guided Practice component.



- What does it mean for two angles to be supplementary?
- Which angles in the quadrilateral are opposite angles?

11.4.3 Guided Practice: Quadrilaterals Inscribed in Circles (continued)

3. Circle R has $VMEH$ inscribed in the circle.
 $m\angle HEM = (2x - 7)^\circ$,
 $m\angle EHV = (2x + 3)^\circ$, and
 $m\angle HVM = (x + 4)^\circ$.



Provide a visual entry point by prompting students to color code the opposite angles in the diagrams.

- a. Find the value of x .
 $x = 61$

- b. Complete the table.

$m\angle HVM = 65^\circ$
$m\angle HEM = 115^\circ$
$m\angle EHV = 125^\circ$
$m\angle EMV = 55^\circ$

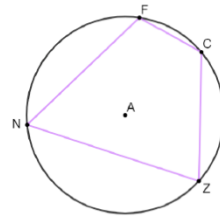
11.4.4 Your Turn!

Component Overview: Students engage in independent practice finding angles in quadrilaterals inscribed in circles. Circulate while students work, to support struggling students and determine which students to call on during the debrief discussion.



11.4.4 Your Turn!

1. Circle A has $NFCZ$ inscribed in the circle. $m\angle NFC = (6x + 32)^\circ$, $m\angle FCZ = (6y - 8)^\circ$, $m\angle NZC = 73^\circ$, and $m\angle FNZ = (3y - 1)^\circ$.



Challenge students who finish early by asking them to construct their own diagram and create problems involving a quadrilateral inscribed in a circle. Students can swap problems with a partner and solve each other's problems.

- a. Find the value of x .

$$x = 12.5$$

- b. What is the $m\angle NFC$?

$$107^\circ$$

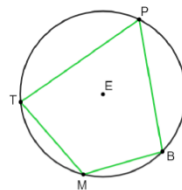
- c. Find the value of y .

$$y = 21$$

- d. What is the $m\angle FCZ$?

$$118^\circ$$

2. Circle E has $TPBM$ inscribed in the circle. $m\widehat{TM} = 71^\circ$, $m\widehat{MB} = 61^\circ$, and $m\angle TMB = (12x - 3)^\circ$.



- a. Find the value of x .

$$x = 9.75$$

- b. What is the $m\angle TMB$?

$$114^\circ$$

11.4.5 Wrap-Up: Error Analysis

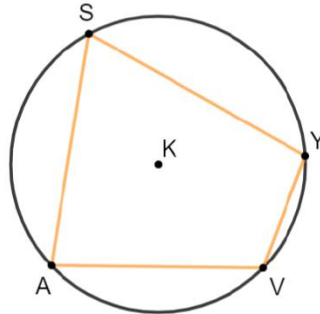
Component Overview: Students engage in analyzing a sample of student work for errors.



11.4.5 Wrap-Up: Error Analysis

1. $SYVA$ is inscribed in Circle K , where
 $m\angle ASY = (3x + 23)^\circ$ and
 $m\angle AVY = (6x + 13)^\circ$.

Anastasia is finding the value of x . Her work is shown.



Students have the opportunity to demonstrate their mastery of the learning target by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors in the future.

Anastasia's Work

$$\begin{aligned} 3x + 23 &= 6x + 13 \\ 3x &= 10 \\ x &= \frac{10}{3} \end{aligned}$$

- a. Write a note to Anastasia explaining her mistake.
Answers will vary. Anastasia, opposite angles of an inscribed quadrilateral are supplementary. To solve for x , add the opposite angles and set them equal to 180° .
- b. Find the value of x .
 $x = 16$